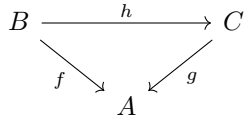
Chapter 9

Slice Category

9.1 Definition

Definition 9.1.1 (Slice Category)**.** For any category $\mathcal{C}$ and $\mathcal{C}$-object A, there is a *slice category* $\mathcal{C}/A$. The objects of $\mathcal{C}/A$ are $\mathcal{C}$-morphisms with codomain A. If there exist $\mathcal{C}$-morphisms $f : B \longrightarrow A$, $g : C \longrightarrow A$, $h : B \longrightarrow C$ and the following triangle commutes, then h is a $\mathcal{C}/A$-morphism with doman f and codomain g. Composition and identity morphisms in $\mathcal{C}/A$ are those of A.

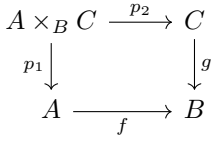


Exercise 9.1.2. Show that if $\mathcal{C}$ has a terminal object 1, then $\mathcal{C}/1$ is equvalent to $\mathcal{C}$.

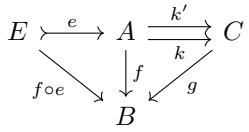
Exercise 9.1.3. Show that, for any category $\mathcal{C}$, each slice $\mathcal{C}/A$ has a terminal object, whether or not $\mathcal{C}$ has a terminal object.

Theorem 9.1.4. If $\mathcal{C}$ has all finite limits, then so does each slice $\mathcal{C}/A$.

Proof. The following pullback is the binary product for $\mathcal{C}/B$-objects $f : A \longrightarrow B$ and $g : C \longrightarrow B$:



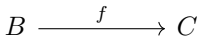
For any $\mathcal{C}/B$-morphisms $k : f \longrightarrow g$ and $k' : f \longrightarrow g$, the following diagram is an equalizer:

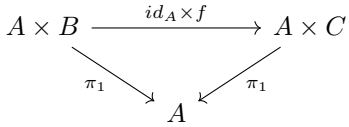


where E, e is an equalizer for k and k' in $\mathcal{C}$. □

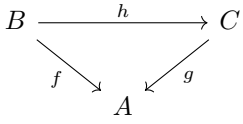
9.2 Slice Functors

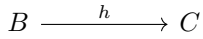
Definition 9.2.1 (Functor A^*)**.** If a category $\mathcal{C}$ has binary products, there is a functor $A^* : \mathcal{C} \longrightarrow \mathcal{C}/A$ for any $\mathcal{C}$-object A, which takes any $\mathcal{C}$-object B to $\pi_1 : A \times B \longrightarrow A$, and any $\mathcal{C}$-morphism $f : B \longrightarrow C$ to $id_A \times f$.





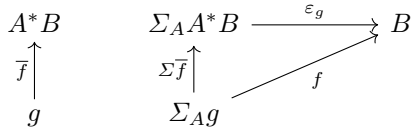
Definition 9.2.2 (Functor Σ_A)**.** For a category $\mathcal{C}$, there is a functor $\Sigma_A : \mathcal{C}/A \longrightarrow \mathcal{C}$ for any $\mathcal{C}$-object A, which takes any $\mathcal{C}/A$ object $f : B \longrightarrow A$ to B, and any $\mathcal{C}/A$ morphism h to itself.



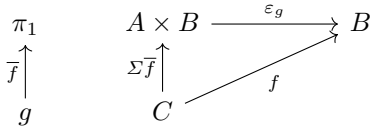


Theorem 9.2.3 ($\Sigma_A \dashv A^*$)**.** If $\mathcal{C}$ has binary products, then for every $\mathcal{C}$ object A, the functor Σ_A is left adjoint to A^*.

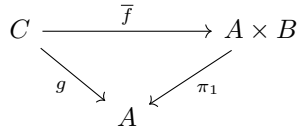
Proof. Consider the following diagram between $\mathcal{C}/A$ and $\mathcal{C}$, where A, B are $\mathcal{C}$-objects and $g : C \longrightarrow A$ is a $\mathcal{C}/A$ object. To show Theorem 9.2.3, it suffices to construct a universal arrow (A^*B, ε_g) from Σ_A to B for all $\mathcal{C}/A$-object g, such that for any $f : \Sigma_A g \longrightarrow B$, there exists a unique adjunct $\overline{f} : g \longrightarrow A^*B$ that makes the following diagram commute:



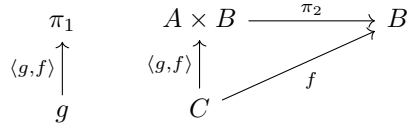
Since $\Sigma_A A^*B = \Sigma_A \pi_1 = A \times B$ and $\Sigma_A g = C$, the above diagram amounts to the following diagram:



where the left diagram in $\mathcal{C}/A$ is the following triangle, which means that $\overline{f}$ must be of the form $\langle g, h\rangle$ for some $h : C \longrightarrow B$.



Let us define the adjunct of f as $\langle g, f\rangle$, which is completely determined by f. Then $\Sigma\overline{f} = \Sigma\langle g, f\rangle = \langle g, f\rangle$. Thus let us further define ε_g as π_2, which makes (π_1, π_2) a universal arrow from from Σ_A to B. The uniqueness of the adjunct is obvious.



By Theorem 6.2.7, η_g is the adjunct of $id_{\Sigma_A g} = id_C$, which is $\langle g, id_C \rangle$. □

Definition 9.2.4 (Internal products)**.** If $\mathcal{C}$ is Cartesian closed, any A-indexed family of objects $f : B \longrightarrow A$ has an A-indexed product $\Pi_A f$ defined by the following pullback.

$$\begin{array}{ccc} \Pi_A f & \longrightarrow & 1 \\ \downarrow & & \downarrow{\scriptstyle \lceil id_A \rceil} \\ B^A & \xrightarrow[f^A]{} & A^A \end{array}$$

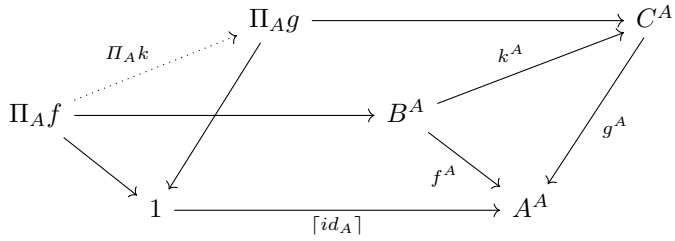
where $\lceil id_A \rceil$ (called the *name of* id_A) is defined as the transpose of $\pi_2 : 1 \times A \longrightarrow A$.

Remark 9.2.5. Intuision behind Definition 9.2.4 is that Π_A is the subset of B^A consisting of internal right-inverses of f. In other words, $\Pi_A f$ is a collection of a A-tuple each of whose element is chosen from each fibre introduced by f.

Definition 9.2.6 (Functor Π_A)**.** If $\mathcal{C}$ is Cartesian closed, there is a functor $\Pi_A : \mathcal{C}/A \longrightarrow \mathcal{C}$ for any $\mathcal{C}$-object A, which takes each $\mathcal{C}/A$-object $f : B \longrightarrow A$ to $\Pi_A f$, and each $\mathcal{C}/A$-morphism $k : f \longrightarrow g$ to $\Pi_A k : \Pi_A f \longrightarrow \Pi_A g$ by pulling the following triangle back along $\lceil id_A \rceil$.

$$\begin{array}{ccc} B^A & \xrightarrow{k^A} & C^A \\ & {\scriptstyle f^A} \searrow \quad \swarrow {\scriptstyle g^A} & \\ & A^A & \end{array}$$

Remark 9.2.7. The existance of $\Pi_A k$ is ensured by Exercise 2.3.6.



Theorem 9.2.8 ($A^* \dashv \Pi_A$)**.** The functor Π_A is right adjoint to A^*.

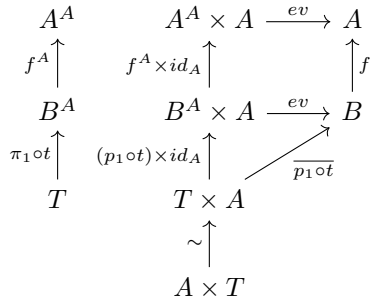
Proof. It suffices to show that there is a functor isomorphism between the comma categories $(A^*, \mathcal{C}/A)$ and $(\mathcal{C}, \Pi_A)$.

Let $f : B \to A$ be a $\mathcal{C}$-morphism. Consider the following series of isomorphisms:

$T \xrightarrow{t} \Pi_A f$	in $\mathcal{C}$
$T \xrightarrow{p_1 \circ t} B^A$	such that $f^A \circ pi_1 \circ t = \ulcorner id_A \urcorner \circ !_T$
$T \times A \xrightarrow{\overline{p_1 \circ t}} B$	such that $f \circ \overline{pi_1 \circ t} = \pi_2$
$A \times T \xrightarrow{\sim} T \times A \xrightarrow{\overline{p_1 \circ t}} B$	such that $f \circ \overline{pi_1 \circ t} = \pi_1$
$A^*T \xrightarrow{(\overline{p_1 \circ t}) \circ \sim} f$	in $\mathcal{C}/\mathrm{A}$

Note that the morphism $\overline{p_1 \circ t} \circ \sim$ is uniquely determined by t. In the third step, we need $f \circ \overline{p_1 \circ t} \circ \sim = \pi_1$ to regard it as $\mathcal{C}/A$ morphism. This holds since:

$$
\begin{array}{ll}
 & f \circ (\overline{p_1 \circ t}) \circ \sim \\
= & ev \circ (f^A \times id_A) \circ (p_1 \circ t \times id_A) \circ \sim \\
= & ev \circ ((f^A \circ p_1 \circ t) \times id_A) \circ \sim \\
= & ev \circ ((\ulcorner idA \urcorner \circ !_T) \times id_A) \circ \sim \\
= & ev \circ (\ulcorner idA \urcorner \times id_A) \circ (!_T \times id_A) \circ \sim \\
= & \pi_2 \circ (!_T \times id_A) \circ \sim \\
= & id_A \circ \pi_2 \circ \sim \\
= & \pi_1
\end{array}
$$



Then we need to show that the right square in $\mathcal{C}/A$ commutes if the left square in $\mathcal{C}$ commutes. This is left as an exercise.

$$
\begin{array}{ccc}
T & \xrightarrow{t} & \Pi_A f \\
{\scriptstyle h}\downarrow & & \downarrow{\scriptstyle \Pi_A k} \\
T' & \xrightarrow[t']{} & \Pi_A g
\end{array}
\qquad
\begin{array}{c} f \\ \downarrow{\scriptstyle k} \\ g \end{array}
\qquad
\begin{array}{c} T \\ {\scriptstyle h}\downarrow \\ T' \end{array}
\qquad
\begin{array}{ccc}
A^*T & \longrightarrow & f \\
{\scriptstyle A^*h}\downarrow & & \downarrow{\scriptstyle k} \\
A^*T' & \longrightarrow & g
\end{array}
$$

The above series of morphisms could be traced from the bottom to the top:

$A^*T \xrightarrow{r} f$	in $\mathcal{C}/A$
$A \times T \xrightarrow{r} B$	in $\mathcal{C}$
$T \times A \xrightarrow{\sim} A \times T \xrightarrow{r} B$	
$T \xrightarrow{\lambda(r\circ\sim)} B^A$	
$T \xrightarrow{!} \Pi_A f$	($\Pi_A f$ is a pullback)

Again, the morphism in the last line is uniquely determined by r. By using this transition, we can show that the right square in $\mathcal{C}$ commutes if the left square in $\mathcal{C}/A$ commutes. This is also left as an exercise.

$$\begin{array}{ccc} T & & \\ {\scriptstyle h}\downarrow & & \\ T' & & \end{array} \qquad \begin{array}{ccc} A^*T & \xrightarrow{\quad r \quad} & f \\ {\scriptstyle A^*h}\downarrow & & \downarrow{\scriptstyle k} \\ A^*T' & \xrightarrow[r']{} & g \end{array} \qquad \begin{array}{ccc} T & \longrightarrow & \Pi_A f \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle \Pi_A k} \\ T' & \longrightarrow & \Pi_A g \end{array} \qquad \begin{array}{c} f \\ \downarrow{\scriptstyle k} \\ g \end{array}$$

□

Exercise 9.2.9. Complete the proof of Theorem 9.2.8.

Theorem 9.2.10. Suppose that a category $\mathcal{C}$ has all finite limits. For each $\mathcal{C}$-object A, if the slice functor A^* has a right adjoint $\Pi_A : \mathcal{C}/A \longrightarrow \mathcal{C}$, then $\mathcal{C}$ is a Cartesian closed.

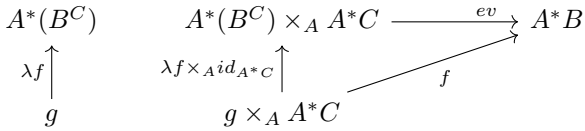
Proof. $_ \times A$ factors as $\Sigma_A \circ A^*$. But $\Sigma_A \dashv A^*$ (by Theorem 9.2.3) and $A^* \dashv \Pi_A$ (by Theorem 9.2.8). By Theorem 6.5.1, $\Sigma_A \circ A^* \dashv \Pi_A \circ A^*$, which gives rise to the exponential. □

Theorem 9.2.11. For any category with binary products, suppose that B^C is an exponential of B by C. Then, for each $\mathcal{A}$-object A, $A^*(B^C)$ is an exponential of A^*B by A^*C in $\mathcal{C}/A$.

Proof. Let $g : C \longrightarrow A$ be a $\mathcal{C}$-morphism. For any $\mathcal{C}/A$-morphism $f : g \times_A A^*C \longrightarrow A^*B$, consider the following series of morphisms:

$g \times_A A^*C \xrightarrow{f} A^*B$	in $\mathcal{C}/A$
$\Sigma_A(g \times_A A^*C) \longrightarrow B$	in $\mathcal{C}$
$(\Sigma_A g) \times C \longrightarrow B$	(By Exercise 2.3.3)
$\Sigma_A g \longrightarrow B^C$	
$g \longrightarrow A^*(B^C)$	in $\mathcal{C}/A$

Let λf be the $\mathcal{C}/A$-morphism in the last line. Then, it suffices to show that the following square commutes:



□

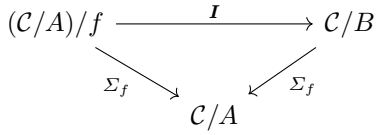
Exercise 9.2.12. Complete the proof of Theorem 9.2.11.

9.3 Functors between Slices

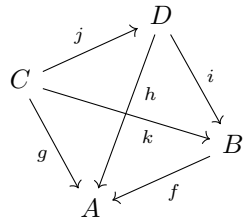
Remark 9.3.1. Let $\mathcal{C}$ be a category and $f : B \longrightarrow A$ is a $\mathcal{C}$-morphism. There is a functor $\Sigma_f : \mathcal{C}/B \longrightarrow \mathcal{C}/A$ that sends each $\mathcal{C}/B$-object $g : C \longrightarrow B$ to $f \circ g : C \longrightarrow A$, and each $\mathcal{C}/B$-morphism $k : g \longrightarrow h$ to $k : f \circ g \longrightarrow f \circ h$.

We use the same symbol Σ_f for this functor as the functor $\Sigma : \mathcal{C}/A/f \longrightarrow \mathcal{C}/A$ but this ambiguity is tolerated by the following isomorphism:

Theorem 9.3.2. For any $\mathcal{C}$-morphism $f : B \longrightarrow A$ there exists a functor isomorphism $\boldsymbol{I}$ that makes the following triangle commute:



Proof. Each $(\mathcal{C}/A)/f$-object is a $\mathcal{C}/A$ morphism with codomain f (ex. $k : g \longrightarrow f$, $i : h \longrightarrow f$), which is also a $\mathcal{C}/B$ object (ex. $k : C \longrightarrow B$, $i : D \longrightarrow B$). Each $\mathcal{C}/B$-object $x : X \longrightarrow B$ is a $\mathcal{C}$-morphism with codomain B, which sends $\mathcal{C}$-morphism $f \circ x : X \longrightarrow A$ to $f : B \longrightarrow A$, thus is a $\mathcal{C}/$A morphism with codomain f.



It is left as an exercise to show that each $(\mathcal{C}/A)/f$-morphism is a $\mathcal{C}/A$-morphism, and each $\mathcal{C}/A$-morphism is a $(\mathcal{C}/A)/f$-morphism. □

Exercise 9.3.3. Complete the proof of Theorem 9.3.2.

Bibliography

Crole, R. L. (1993) *Categories for Types*. Cambridge University Press.

Kleisli, H. (1965) "Every standard construction is induced by a pair of adjoint functors", In the Proceedings of *the American Mathematical Society*, Vol. 16. pp.544–546.

MacLane, S. (1997) *Categories for the Working Mathematician*, Graduate Texts in Mathematics. Springer, second edition edition.

MacLane, S. and I. Moerdijk. (1992) *Sheaves in Geometry and Logic – A First Introduction to Topos Theory –*. Springer.

McLarty, C. (1992) *Elementary Categories, Elementary Toposes*, Oxford Logic Guides 21. Oxford Science Publications.

Moggi, E. (1989) "Computational lambda-calculus and monads", In the Proceedings of *Fourth Annual IEEE Symposium on Logic in Computer Science*. pp.14–23.

Pedicchio, M. C. and W. Tholen. (2004) *Categorical Foundations: Special Topics in Order, Topology, Algebra, and Sheaf Theory*, Vol. 97 of *Encyclopedia of Mathematics and Its Applications*. Cambridge University Press.